

Boundary Case: Single-Cell Deployment

The Minimum Viable Architecture Where One Cell, One Aspect, and One Self Constitute the Complete Deployment, Testing Governance Architecture at Minimum Scale While Remaining Architecturally Valid

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

The Coordination Knowledge Substrate (CKS) architecture specifies no minimum cell count. A deployment consisting of exactly one cell, organized into one aspect, integrated into one Self is architecturally valid: all three structural levels per B1.02 are present, all Paper 2 governance commitments apply, and lifecycle and evolution governance operate without modification. This note formalizes the single-cell deployment as a boundary case — the minimum viable configuration — and examines what the architecture’s commitments mean when tested against their lower bound. The architectural boundaries under examination are: aspect coordination with one cell member (B1.04), productive tension with minimal operational evidence (B1.12), and content-domain composition at minimum scale (B1.18). The governance implications are real but differently weighted than at multi-cell scale: the aspect provides genuine governance value even with one member, the three-level structure is present and operable, and all lifecycle commitments apply. Three stress points require active monitoring: scale inversion risk toward Monolithic Cell (B3.09) if the single cell grows without decomposition, Purposeless Aspect drift (B3.05, Form 2) if the aspect never acquires a coherent expansion purpose, and evidence poverty in action-feedback when execution volume is minimal. The single-cell deployment validates that the architecture applies at any scale while also naming where some governance features provide minimal operational benefit at this scale.

1. Configuration Description

The single-cell deployment is the minimum viable CKS configuration. It consists of exactly one cell executing one specific informational task; that cell is the sole member of one aspect; that aspect is the

sole aspect of one Self. Nothing about this configuration violates any architectural commitment in Paper 2 (Li, April 2026). The three structural levels required by B1.02 are present: cell, aspect, Self each exist as distinct structural entities with distinct governance responsibilities. All Paper 2 commitments apply without modification.

The cell carries the full Paper 1 (Li, April 2026) architecture: a substrate, orchestration rules, the six Paper 1 commitments, a DNA layer containing stabilized orchestration substrates and behavior substrates, and an action layer accumulating recorded task instances. The aspect has a purpose statement (B2.15) specifying the coordination purpose it exists to serve, and coordination rules (B2.16) governing how the cell contributes to that purpose. The Self is the integrated whole holding the single aspect as the sole facet of one CKS-governed intelligence.

This configuration is correctly understood as minimum viable rather than as degenerate or deficient. The governance infrastructure is complete. The structural levels are populated. The architecture's commitments hold at every layer. What distinguishes the single-cell deployment from multi-cell deployments is not an architectural deficiency but a scale condition: some governance features that provide meaningful operational benefit when multiple cells and aspects are present provide minimal operational benefit when only one of each exists. The architectural limits section (§6) distinguishes these features explicitly.

2. Architectural Boundaries Being Tested

Three architectural boundaries from Paper 2 are tested at minimum scale.

B1.04 — Aspect coordination with one cell member. An aspect is a coordination arrangement of cells serving a particular purpose, operating over its constituent cells as content domain. At multi-cell scale, this means the aspect coordinates contributions from multiple cells, asks pattern questions across them, and integrates their outputs toward the aspect's purpose. With one cell, there are no peers to coordinate. The boundary question is: what does coordination mean when there is nothing to coordinate across? The answer the architecture provides is that the aspect's coordination function is degenerate at minimum scale — not absent. The aspect's purpose statement still defines what informational territory the aspect governs. Its coordination rules still specify how the single cell contributes to that purpose. The coordination behavior has no second party, but the coordination governance is fully present. The aspect exists not for the single-cell coordination it cannot perform but for the governance function and classification function it provides regardless of member count.

B1.12 — Productive tension at minimum scale. Productive tension in the CKS architecture arises from the interaction of evolution mechanisms operating on the same entities: mutation (instinct evolution) modifying the LLM layer, directed selection (B1.14) refining cell DNA, and action-feedback (B1.15) closing the loop from recorded execution back toward DNA improvement. At multi-cell scale, horizontal evolution (B2.80) propagates improvements from one cell's DNA to peer cells, creating cross-cell productive tension. With one cell, the horizontal axis has no peers to

propagate to. Mutation and directed selection still apply — the single cell’s LLM layer can be updated and its DNA can be improved — but horizontal propagation produces no downstream effect. Action-feedback operates but on whatever evidence the single cell’s execution history provides, which at minimum scale may be sparse. The productive tension that does exist is intra-cell rather than cross-cell: DNA evolution tension, the interaction between instinct and reasoning evolution within one cell. The cross-cell productive tension that makes horizontal evolution architecturally interesting is minimal.

B1.18 — Content-domain composition at minimum scale. Composition compatibility (B2.92) between a cell and an aspect requires that the cell’s informational domain fits within the aspect’s coordination purpose, and that the aspect’s purpose is coherent across its member cells. With one cell, composition compatibility reduces to a binary: either the single cell covers the aspect’s purpose entirely, or it does not. There are no partial-coverage questions, no cross-cell domain boundary questions, no coordination seam questions. Composition is either satisfied (the cell fits the aspect’s purpose) or not satisfied (the cell is in the wrong aspect or the aspect’s purpose is misspecified). The composition governance is fully present; it is simply undemanding at minimum scale.

3. Governance Implications

Single-member aspect governance. The aspect with one cell member still requires all aspect-level governance. A purpose statement (B2.15) must exist: what informational territory does this aspect govern? Coordination rules (B2.16) must exist: how does the cell contribute to the aspect’s purpose? The governance infrastructure is complete. The coordination behavior that infrastructure supports is minimal — because there is one cell and no peers to coordinate — but minimality of behavior is not the same as absence of governance. A human with appropriate authority can inspect, modify, and override the aspect’s purpose statement and coordination rules at any time, which is the CKS governance commitment, and that commitment holds whether the aspect has one member or twenty.

Aspect value at minimum scale. The aspect provides four governance values even with one cell. First, classification: the aspect marks the cell’s operational territory as aspect-governed with a specified purpose. Second, governance-layer provision: the aspect provides the governance coordination layer that enables future cell additions without architectural change. Adding a second cell to the aspect does not require a new architecture; it requires assigning a new cell to the existing governance structure. The aspect’s existence at minimum scale is what makes the deployment expansion-ready. Third, three-level structure maintenance: the aspect preserves the three-level structure (B1.02) that Paper 2’s architectural commitments are built on. Collapsing the aspect would reduce the deployment to a two-level structure, which is Paper 1 architecture, not Paper 2 architecture. Fourth, future composition compatibility: an aspect with a coherent purpose statement can be composed with other aspects within the Self as the Self grows. The governance work done at minimum scale — writing the purpose statement, defining coordination rules — is the governance work that makes future composition possible.

Lifecycle governance at minimum scale. Birth governance (B1.09) applies to all three entities: the cell, the aspect, and the Self are each governed into existence through the birth commitment. Death governance (B1.11) applies if any entity is closed. Mating (B1.10) applies if the single cell produces offspring: the mating pattern must be selected under governance and the offspring's provenance preserved. All lifecycle commitments operate at full force regardless of scale.

Evolution at minimum scale. Mutation governance (B1.13) applies: LLM updates that affect the single cell's instinct layer require the verification and governance commitments that mutation governance specifies. Directed selection (B1.14) applies: the single cell's DNA can be improved through human-governed substrate editing. Action-feedback (B1.15) operates: recorded execution instances from the single cell feed into proposals for DNA improvement, subject to human mediation before taking effect. Horizontal evolution (B2.80) has no peers to propagate to — the mechanism exists architecturally but produces no cross-cell effect. Vertical evolution operates in its structural-reorganization sense: the single-aspect arrangement can be reorganized through governed substrate editing.

4. Boundary Tests

Three boundary tests apply to the single-cell deployment. Each names a condition the deployment must satisfy to remain architecturally valid.

(a) Purpose statement survival under expansion. Does the single aspect's purpose statement (B2.15) articulate a coordination purpose that would remain coherent if a second cell were added to the aspect? A purpose statement that only makes sense for exactly one cell — whose content is indistinguishable from a description of the single cell's behavior — is not a coordination purpose statement; it is a cell description masquerading as an aspect purpose. A valid aspect purpose statement names what the aspect is coordinating for, not merely what the cell it contains does. The operational test: if a second cell were added to this aspect tomorrow, would the existing purpose statement still govern the new cell's contribution? If the purpose statement would need to be entirely rewritten to accommodate a second cell, the aspect may be approaching Purposeless Aspect (B3.05, Form 2 — pass-through aspect with no durable coordination purpose).

(b) Action-feedback quality at minimal evidence volume. Does the action-feedback mechanism (B1.15) produce useful DNA improvement proposals from the minimal execution history a single cell at minimum scale provides? At minimum scale, the action layer may contain few recorded instances. Sparse evidence can produce proposals that are statistically premature — overfitting to a small sample. The governance architecture handles this through the human mediation requirement: action-feedback proposals are proposals, not automatic changes, and the human who authorizes integration should calibrate their authority accordingly. Proposing substrates should be configured to signal evidence volume alongside proposals, so that human mediators can evaluate proposals with appropriate epistemic weight given the execution history size.

(c) Three-level structure governance value. Does the three-level structure add governance value at minimum scale beyond what a single-level deployment would provide? It adds governance value in the form of the classification, expansion-readiness, and lifecycle governance named in §3, but minimal operational complexity management value at single-cell scale. If the governance obligations the structure imposes are not being discharged — if the aspect has no real purpose statement, if the Self has no governed coherence beyond the single cell — the deployment is nominally Paper 2 but substantively Paper 1 with unused structure layered on top.

5. Stress Points

Scale inversion risk — Monolithic Cell (B3.09). The single-cell deployment becomes a scale-inversion failure if the single cell is permitted to grow in scope without being decomposed. CKS architecture decomposes large, complex tasks into multiple cells coordinated by one or more aspects. If a single cell is used as a container for what should be multiple cells — executing across multiple informational territories, carrying DNA that governs behaviors spanning multiple distinct purposes — the single-cell deployment is not a minimum viable configuration; it is a Monolithic Cell (B3.09) anti-pattern. The signal is not cell size per se but cell scope: a cell whose DNA governs work across informationally distinct territories, which could coherently be split into two or more cells each with their own aspect membership. Governance response: when the single cell's scope expands, evaluate whether the expansion represents organic growth within the cell's informational territory or scope creep that should be addressed by adding a second cell and potentially a second aspect.

Purposeless Aspect drift — B3.05, Form 2. A single-member aspect that never acquires additional members and whose purpose statement drifts toward describing the single cell's behavior rather than naming a genuine coordination purpose is approaching the Purposeless Aspect anti-pattern (B3.05), specifically Form 2: a pass-through aspect that exists structurally but provides no coordination governance of its own. The risk at single-cell scale is not that the aspect has one member (one-member aspects are architecturally valid) but that the governance work the aspect requires — maintaining a coherent purpose statement, maintaining coordination rules — is not being performed. Governance response: the aspect's purpose statement and coordination rules should be reviewed when the deployment is evaluated for growth. A purpose statement that cannot answer “what would a second cell in this aspect need to do to satisfy the aspect's purpose?” is a candidate for respecification.

Evidence poverty — action-feedback calibration. At minimum scale, the single cell's action layer may not accumulate sufficient execution instances for action-feedback (B1.15) to produce well-evidenced DNA improvement proposals. This is not an architectural failure but a calibration challenge: the mechanism operates correctly, but its outputs carry less evidential weight than they would at higher execution volumes. Governance response: action-feedback proposing substrates at minimum scale should surface evidence volume as part of the proposal, and human mediators should apply appropriate skepticism to proposals generated from small execution samples. DNA evolution

(B1.13) through directed selection may be more reliable than action-feedback-driven evolution at minimum scale until the action layer accumulates sufficient depth.

6. Architectural Limits

The CKS architecture specifies no minimum cell count. A single-cell deployment is architecturally valid, and the governance implications described in §3 are those that apply. Nothing in Paper 2 (Li, April 2026) requires a deployment to have more than one cell, more than one aspect, or more than one level of aspect composition. The minimum viable configuration described in this note satisfies all structural requirements.

However, the architecture is designed for multi-cell environments, and naming this honestly is not a disqualification of minimum-scale deployments but an accurate statement of where the architecture's features provide maximum benefit. The following features provide minimal operational benefit at single-cell scale: horizontal evolution (B2.80), since peer-to-peer DNA propagation produces no effect with one cell; aspect-level conflict handling, since there are no inter-cell conflicts within the aspect to handle; and multi-shaped governance complexity, since governance across all three levels may be exercised by the same person at minimum scale without governance loss.

These limitations are features of scale, not architectural deficiencies. They identify where the architecture's investment pays off most clearly and where a deployment that will remain permanently at minimum scale may be more simply operated as a Paper 1 deployment. The architectural conclusion is this: single-cell deployment is the minimum viable configuration under Paper 2, validates that Paper 2's commitments apply at any scale, and establishes the governance foundation that makes expansion from minimum scale to full multi-cell deployment possible without architectural surgery. The value of beginning with a Paper 2 architecture at minimum scale is forward-compatibility: every governance decision made at minimum scale is a decision that will still hold when the deployment grows.

7. Conclusion

The single-cell deployment demonstrates that the CKS architecture is scale-complete: its commitments apply without modification from one cell upward, and the governance infrastructure they require is fully instantiable at minimum scale. The three-level structure is not a multi-cell requirement; it is a structural requirement that happens to be minimally populated at single-cell scale. All lifecycle and evolution governance operates. All human-governance commitments hold.

The boundary the single-cell deployment marks is not architectural validity — which is unconditional — but operational density. Some governance features provide rich operational benefit when multiple cells and aspects are present, and minimal operational benefit when only one cell exists. Naming this distinction is not a critique of minimum-scale deployments; it is an accurate description of what the architecture's features are for and where they deliver their value.

The governance obligations the three-level structure imposes at minimum scale — maintaining a genuine aspect purpose statement, monitoring for scale inversion risk, calibrating action-feedback to evidence volume — are the same governance obligations that make the deployment growth-ready. A single-cell deployment that discharges its governance obligations correctly is not a simplified deployment; it is a correctly governed deployment at minimum scale, from which the full CKS architecture can grow without architectural change.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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